



Wildfire Detection System

Sponsored by Whatcom Million Trees

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MAKE WAVES.

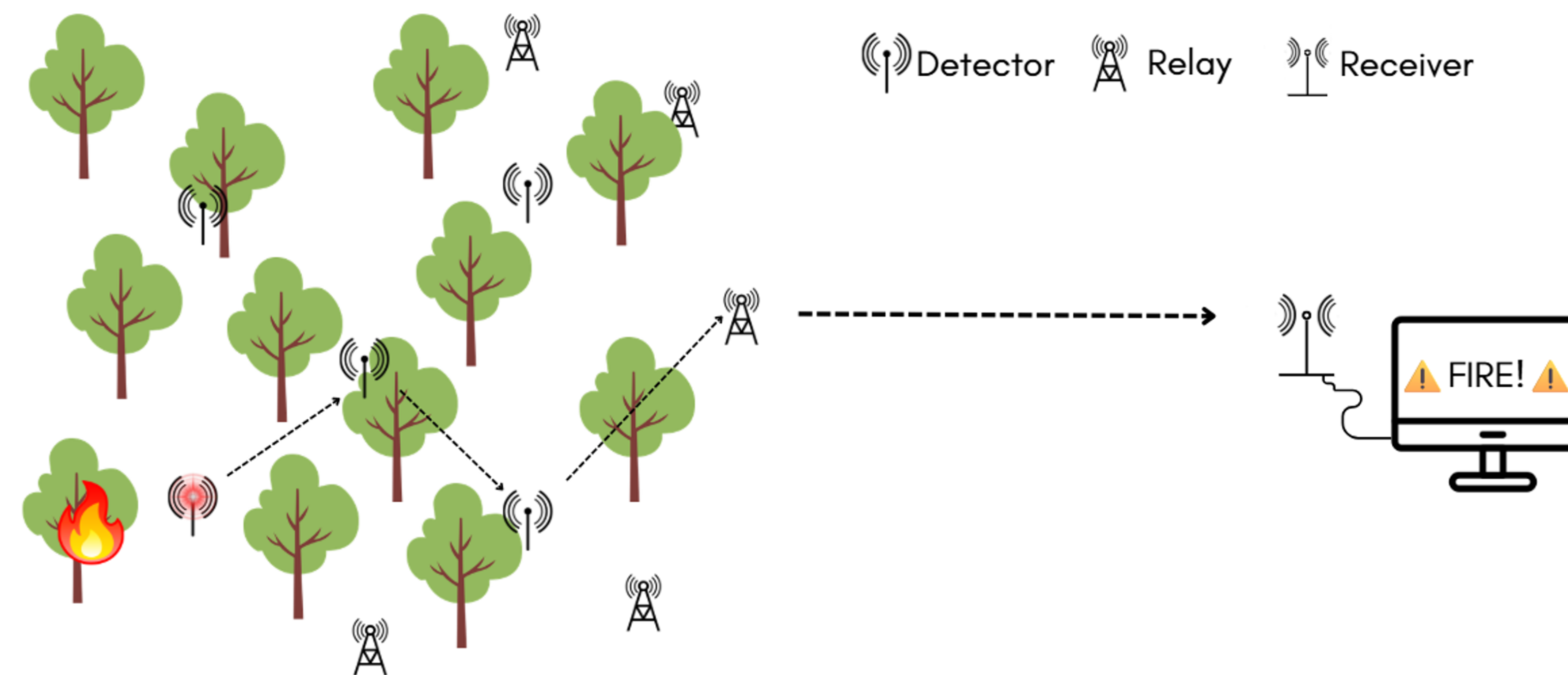
Abstract

Since the 1980s, wildfires have been increasingly common, with a notable increase in severity and amount of acreage burned. Wildfires are more common in areas such as Australia, but with the rise in global temperatures, Washington has also seen a rise in wildfires particularly during lightning season but with a rise of wildfires in the summer. Many forests in Washington state are very dense with large trees or are near residential areas. This makes it difficult and dangerous for firefighters to access and obtain accurate alerts. Our system is made to mitigate the damage from wildfires using low-power wildfire detectors, which provide early alerts to fire and forest authorities in the event of a large-scale fire as well as monitoring environmental conditions to identify signs of wildfire activity using machine learning and transmit real-time alerts to improve emergency response times. The aspect of low power consumption will enable the sensor to be easily deployed in remote areas with minimal maintenance.

Challenges Encountered

- Lack of wildfire data
- Hardware shipping delays
- SPI inconsistencies
- Lack of documentation for LoRa module
- Unofficial test board for BME
- No market battery holder
- Battery charging and spot-welding
- No break out board for LoRa and fuel gauge

Algorithm and system design

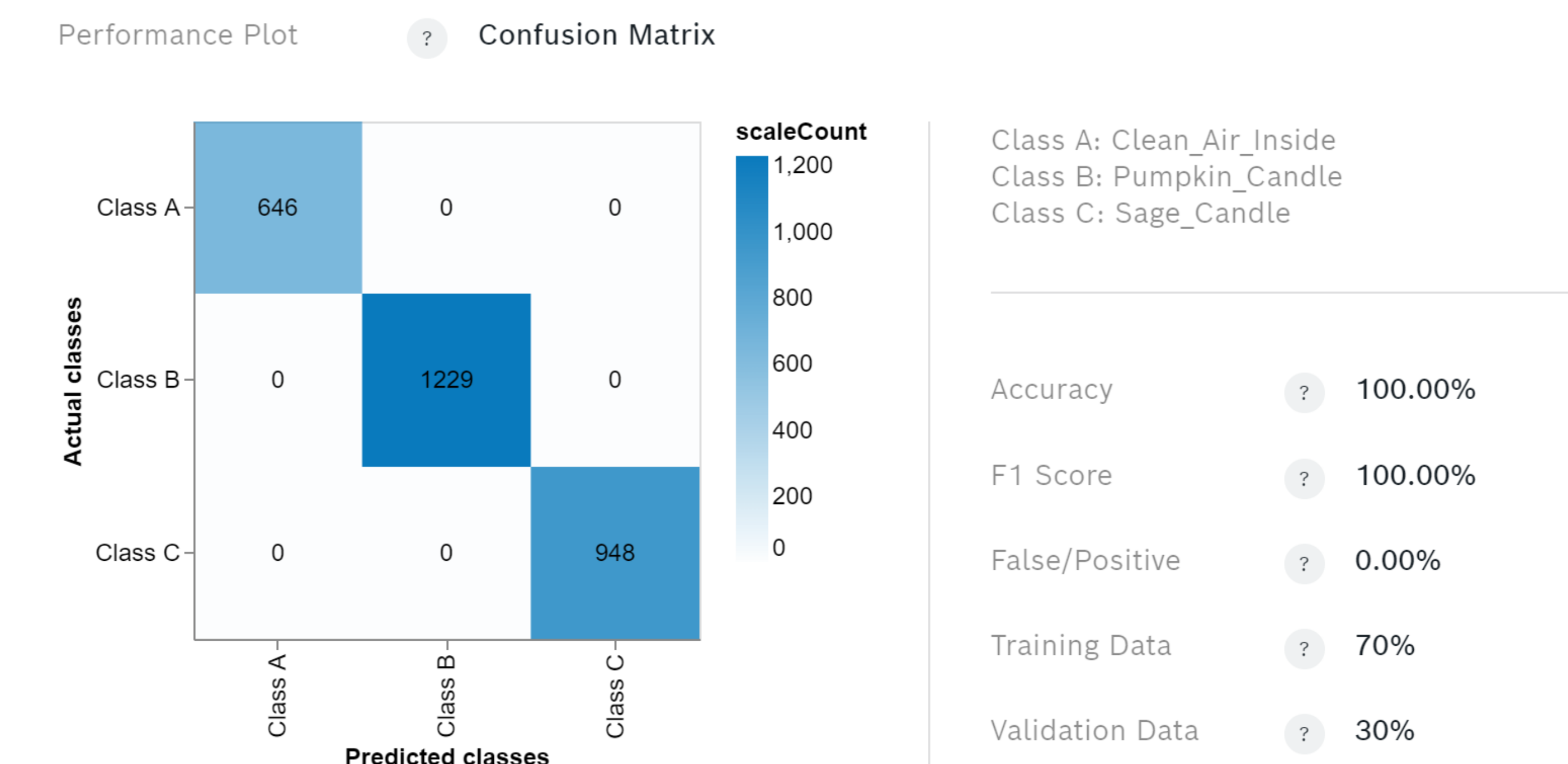


Above: visual demonstration of system in forest

Detectors are set up in a grid throughout a forest to create a mesh network where any wildfire in the forest will be detected and an alert will bounce between detectors until a relay can send a long-distance message to alert wildfire authorities.

Summary of Results

- We have successfully created the groundwork for low power wildfire detection system.
- Each detector has an estimated battery life of 5 years.
- Average detection time of 1 minute at a range of up to 5 meters.
- Lack of data drives amount of false positives
- Training is possible for wildfires, barbecues, cigarettes, and other nicotine products.
- Inconsistencies over time with more exposure to candles.
- Can perform in high temperatures.
- Mesh-network support.
- Easy-to-read alert data



Above: BME AI studio gathered candle data to create machine learning algorithm to output a configuration file to our final product. This file allows our detector to identify wildfires and distinguish between other classes like cigarettes and smaller campfires.

Conclusion

While our AI detection system is trained on candles, an implementation into forests with wildfire data is a very real possibility. Many scented candles contain the same chemical signature as wildfires meaning accurate detection of candles is comparable to wildfires. For implementation in forests, fire data should be collected based on forest type.

Acknowledgements

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Capstone Project
Electrical & Computer Engineering Program